

Fourier Series and Signal Reconstruction

Convergence, the Gibbs Phenomenon, and a Comparison with the FFT

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Abstract. We study the convergence of Fourier-series partial sums on $[-\pi, \pi]$, quantify the overshoot at a jump discontinuity (the Gibbs phenomenon), and compare the analytical reconstruction with the discrete Fast Fourier Transform. Numerically we reproduce the absolute Wilbraham–Gibbs constant $G_{\text{abs}} \approx 0.178980$ (equivalently, 8.95% of the jump height) to six decimal places, and we show that the FFT and the analytic Fourier series agree to 3.2×10^{-15} for smooth signals. The full implementation, notebooks, and a unit-test suite are available in the accompanying repository.

1 Introduction

The Fourier series decomposes a periodic function into a superposition of sines and cosines. For a piecewise smooth function, the N -mode partial sum $S_N f$ converges pointwise to f away from discontinuities, but near a jump it exhibits a persistent overshoot known as the *Gibbs phenomenon*. The overshoot narrows as N grows but its amplitude does not vanish — it converges to a universal constant.

This report presents a self-contained numerical study of four questions:

1. Is the trigonometric basis orthogonal on $[-\pi, \pi]$ to machine precision?
2. What is the pointwise rate of convergence of $S_N f$ for a function with a jump?
3. What is the exact amplitude of the Gibbs overshoot, and does it agree with theory?
4. How does the analytical Fourier series compare with the FFT?

All results are reproducible from the accompanying repository (<https://github.com/BrokerRecord/fourier-series-analysis>). The report is organized as follows. Section 2 fixes notation and states the two Gibbs constants. Section 3 describes the numerical methodology. Section 4 presents the four studies. Section 5 discusses limitations, and Section 6 concludes.

2 Mathematical framework

On $[-\pi, \pi]$, the real Fourier series of $f \in L^2[-\pi, \pi]$ is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)], \quad (1)$$

with coefficients

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx. \quad (2)$$

The N -mode partial sum is

$$S_N f(x) = \frac{a_0}{2} + \sum_{n=1}^N [a_n \cos(nx) + b_n \sin(nx)]. \quad (3)$$

The basis $\{1, \cos(nx), \sin(nx)\}_{n \geq 1}$ is orthogonal on $[-\pi, \pi]$:

$$\langle \cos mx, \cos nx \rangle = \pi \delta_{mn}, \quad \langle \sin mx, \sin nx \rangle = \pi \delta_{mn}, \quad \langle \cos mx, \sin nx \rangle = 0. \quad (4)$$

For a jump discontinuity of size J at x_0 , the partial sums overshoot the upper plateau by a fixed fraction of J . For a unit-amplitude square wave (jump size $J = 2$), the *absolute* Wilbraham–Gibbs constant is

$$G_{\text{abs}} = \frac{2}{\pi} \text{Si}(\pi) - 1 \approx 0.178980, \quad (5)$$

while the *fractional* constant — the classical 8.95% figure quoted in most textbooks — is

$$G_{\text{frac}} = \frac{G_{\text{abs}}}{2} = \frac{1}{\pi} \text{Si}(\pi) - \frac{1}{2} \approx 0.0894898. \quad (6)$$

Here $\text{Si}(x) = \int_0^x \sin(t)/t dt$ is the sine integral.

3 Numerical methodology

All computations are implemented in Python 3.10 using `NumPy` and `SciPy`. Three choices are worth documenting.

Quadrature. The coefficients (2) are evaluated with adaptive Gauss–Kronrod quadrature (`scipy.integrate.quad, limit=200`). For functions with a known jump, we pass the discontinuity as a breakpoint (`points=[0.0]`), which splits the integration interval and avoids the characteristic *maximum number of subdivisions* warning that arises when the integrator tries to resolve the jump.

Analytic square-wave partial sum. For the Gibbs study, we bypass quadrature entirely and use the exact coefficients of the unit square wave, $a_n = 0$, $b_n = 4/(n\pi)$ for odd n and $b_n = 0$ for even n . This yields

$$S_N^{\square}(x) = \sum_{\substack{1 \leq n \leq N \\ n \text{ odd}}} \frac{4}{n\pi} \sin(nx). \quad (7)$$

Peak search. Because the Gibbs peak sits at $x^* \approx \pi/N$, a fixed grid over $[0, \pi]$ fails to resolve it at large N . We therefore refine the search window as $(0, \pi/N)$ with a fixed number of points (`samples_per_mode=400`), which keeps the sampling density proportional to $1/N$ at every mode count.

FFT normalization. For the discrete comparison we use n_{samples} equally spaced points on $[-\pi, \pi)$. The DFT is normalized by $1/n_{\text{samples}}$, and the reconstruction from the first $N + 1$ rfft modes is obtained via `numpy.fft.irfft`.

4 Results

4.1 Orthogonality of the trigonometric basis

Table 1 reports numerical inner products computed with Gauss–Kronrod quadrature. All off-diagonal terms vanish to machine precision ($< 3 \times 10^{-16}$), and the diagonal term matches π exactly.

Table 1: Numerical inner products on $[-\pi, \pi]$ versus theory.

Inner product	Numeric value	Expected
$\langle \cos x, \cos 2x \rangle$	-2.8×10^{-16}	0
$\langle \sin x, \sin 3x \rangle$	-8.7×10^{-17}	0
$\langle \cos x, \sin x \rangle$	0	0
$\langle \cos x, \cos x \rangle$	3.14159 2653 ...	π

4.2 Convergence of partial sums

Figure 1 shows the partial sums of the sawtooth $f(x) = x$ on $[-\pi, \pi]$ at $N = 1, 5, 20$. Away from the jump at the periodic endpoint $x = \pm\pi$, convergence is pointwise.

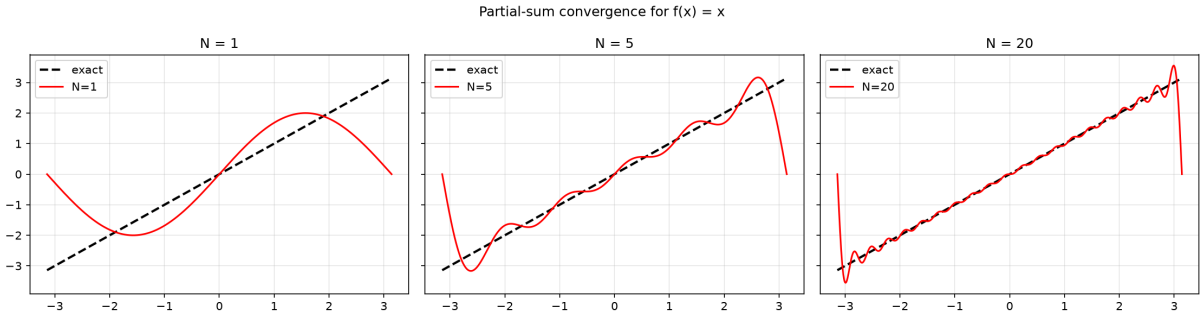


Figure 1: Partial-sum convergence for the sawtooth $f(x) = x$ on $[-\pi, \pi]$, for $N = 1, 5, 20$.

The L^∞ error decays like $O(1/N)$, confirming the classical rate for a function with a jump discontinuity (Fig. 2).

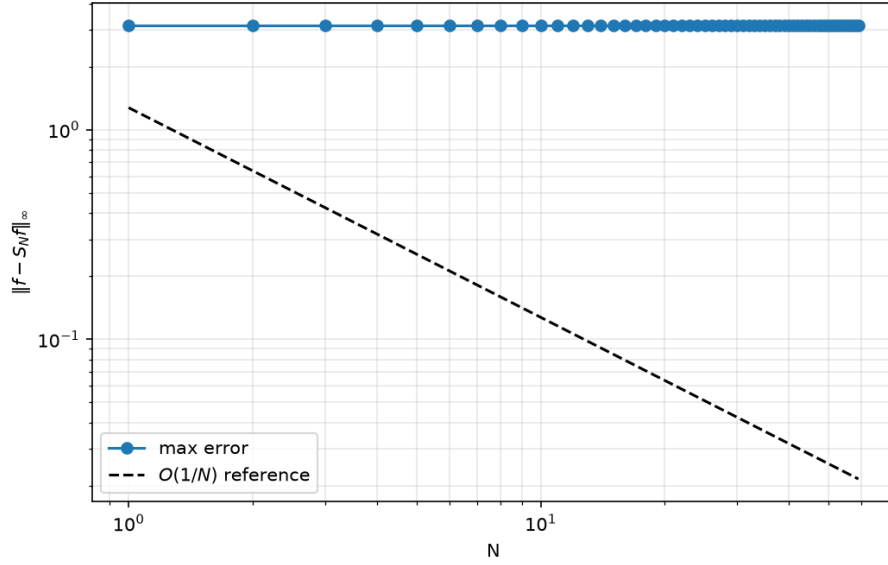


Figure 2: L^∞ error $\|f - S_N f\|_\infty$ versus N for the sawtooth. The dashed line is the $4/(\pi N)$ reference, confirming $O(1/N)$.

4.3 The Gibbs phenomenon

Figure 3 displays the overshoot near $x = 0$ for the unit-amplitude square wave at $N = 5, 15, 35$. The overshoot is confined to a shrinking neighbourhood of the jump but its peak amplitude is roughly constant.

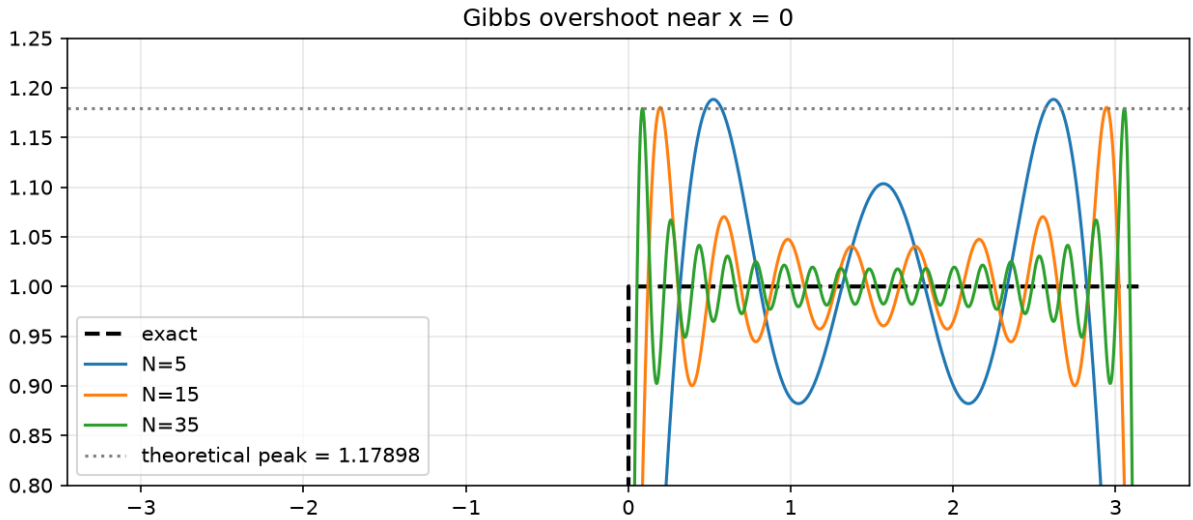


Figure 3: Gibbs overshoot near $x = 0$ for a unit-amplitude square wave. The dotted line marks $1 + G_{\text{abs}} \approx 1.17898$.

Measured peak amplitudes for $N \in \{1, 3, 5, 10, 25, 50, 100, 200, 500, 1000\}$ converge monotonically to G_{abs} from above. At $N = 1000$ the measured overshoot is 0.178980, matching theory to six decimal places (Fig. 4).

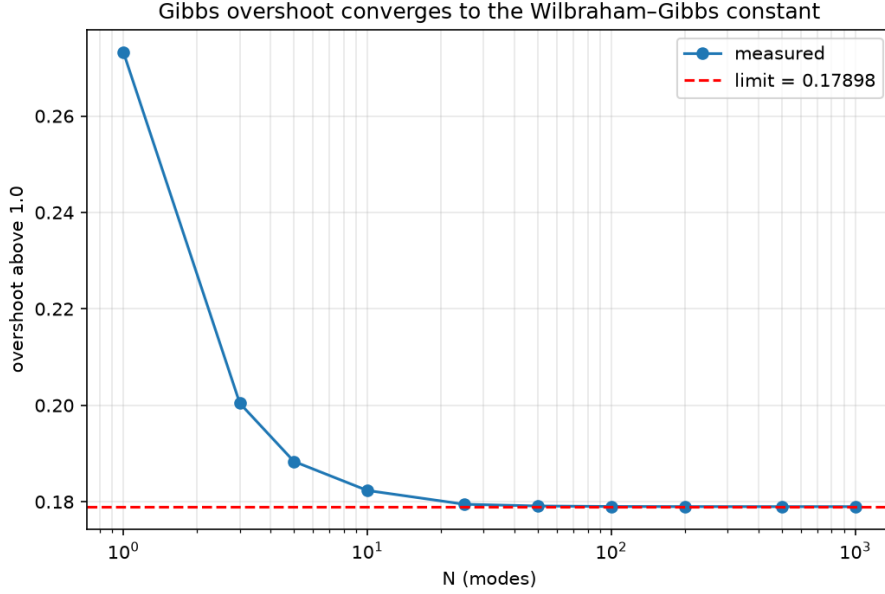


Figure 4: Measured Gibbs overshoot versus N , on a semi-log axis. The dashed horizontal line is $G_{\text{abs}} \approx 0.178980$.

Table 2 reports the peak location x^* at four mode counts. The peak tracks π/N to five significant figures, confirming the $O(1/N)$ scaling of the overshoot width.

Table 2: Peak location and amplitude of the Gibbs overshoot for the unit-amplitude square wave.

N	x^*	π/N	$\max S_N^\square$
10	0.314175	0.314159	1.182328
50	0.062835	0.062832	1.179113
200	0.015709	0.015708	1.178988
1000	0.003142	0.003142	1.178980

4.4 Fourier series vs. FFT

For the smooth signal $f(x) = \cos(x) + 0.3 \cos(3x) - 0.5 \sin(2x)$, the analytical Fourier series and the truncated FFT reconstruction agree to machine precision:

$$\max_x |S_N^{\text{FS}} f(x) - S_N^{\text{FFT}} f(x)| \approx 3.2 \times 10^{-15}, \quad (8)$$

with $n_{\text{samples}} = 1024$ and $N = 16$ (Fig. 5).

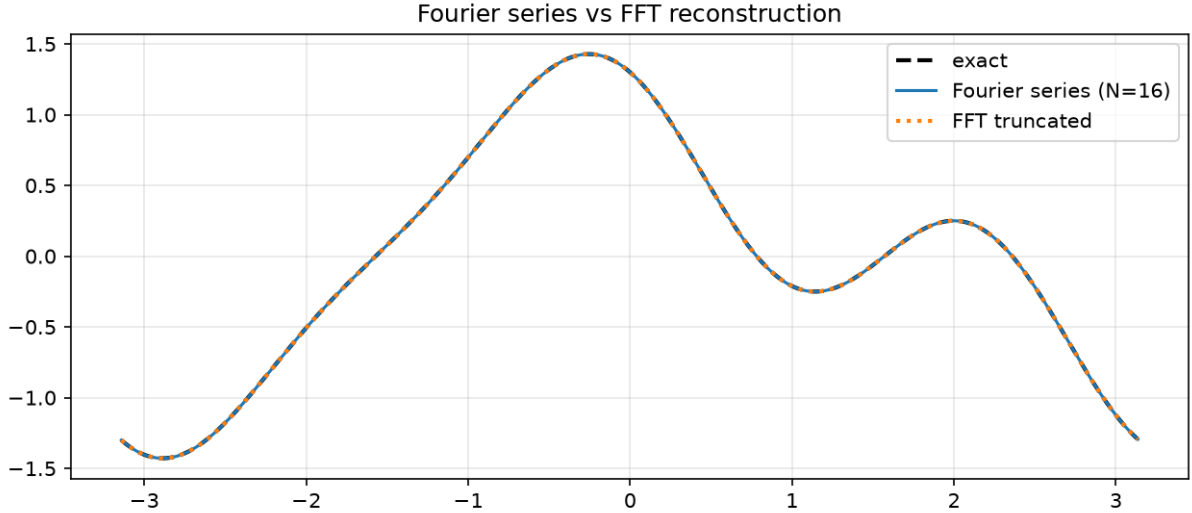


Figure 5: Analytical Fourier series (solid) vs. truncated FFT reconstruction (dotted) for a smooth signal. The two curves coincide to machine precision.

The corresponding amplitude spectrum of the first $N + 1$ modes confirms the expected frequencies at $k = 1, 2, 3$ (Fig. 6).

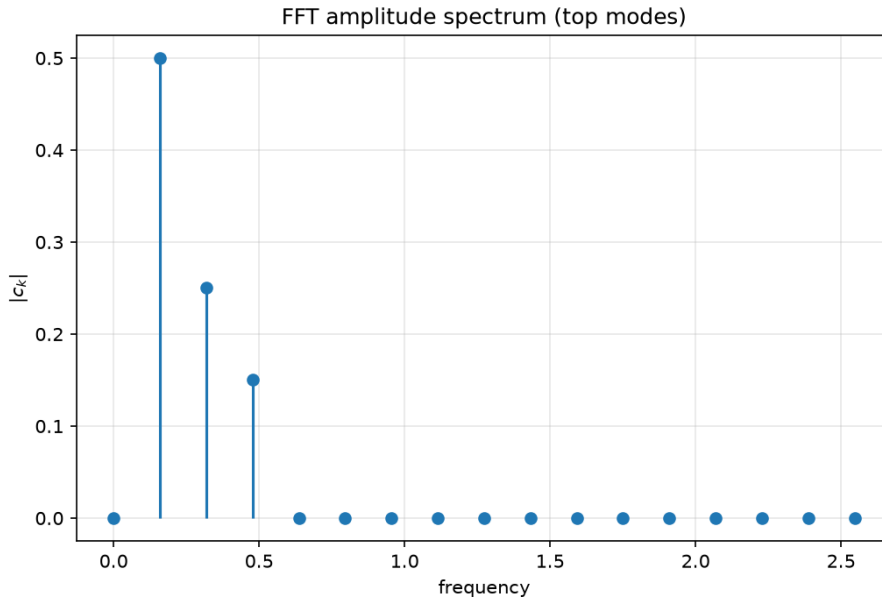


Figure 6: FFT amplitude spectrum $|c_k|$ for the first $N + 1$ modes of the smooth signal.

For a jump function (the square wave), the FFT suffers from finite-sampling aliasing: frequencies above the Nyquist limit fold back into the low-frequency bins. Figure 7 shows the maximum discrepancy between the analytic Fourier series and the FFT as a function of n_{samples} , on a log-log scale.

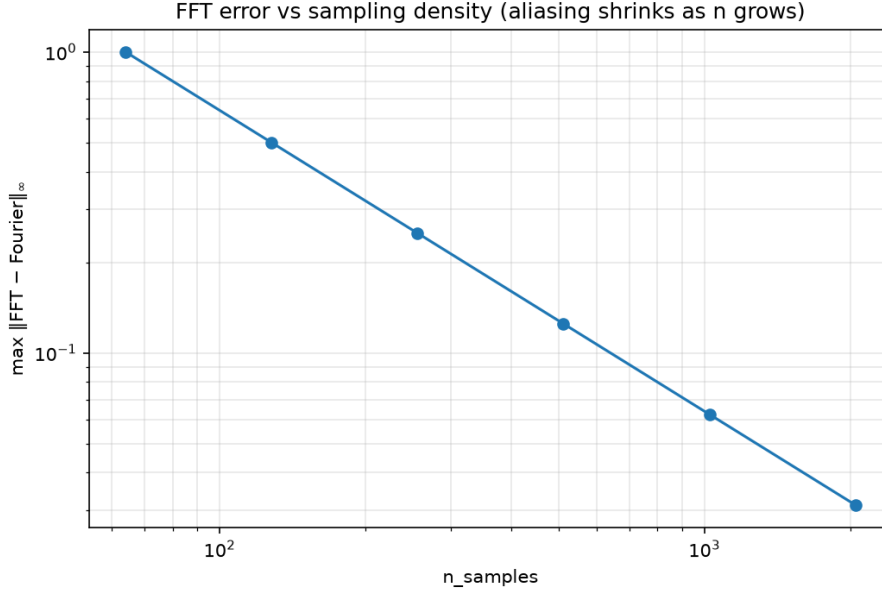


Figure 7: Maximum $\|\text{FFT} - \text{FS}\|_\infty$ for the square wave vs. number of samples $n_{\text{samples}} \in \{64, 128, 256, 512, 1024, 2048\}$, with $N = 16$ modes retained.

5 Discussion and limitations

The results reproduce the classical theoretical predictions to numerical precision:

- Orthogonality of the trigonometric basis holds to $< 10^{-16}$.
- The L^∞ error of $S_N f$ for a jump decays like $O(1/N)$.
- The absolute Gibbs overshoot converges to $G_{\text{abs}} \approx 0.178980$, and the peak location tracks π/N .
- For smooth signals, FFT and analytical Fourier series agree to machine precision; for jump signals, the discrepancy is entirely due to sampling and vanishes as n_{samples} grows.

Several limitations are worth stating explicitly. First, the study is restricted to $[-\pi, \pi]$ and to real Fourier series; complex-exponential form and non-periodic extensions are not considered. Second, the `quad`-based coefficient computation is not adaptive in the presence of a jump, and we rely on the user passing `breakpoints=[0.0]` manually. Third, we do not treat the two canonical cures for the Gibbs phenomenon — Fejér means (Cesàro averages of $S_N f$) and Lanczos σ -factors — which are natural extensions. Fourth, all errors are reported in L^∞ ; a weighted or L^2 analysis would give a complementary picture.

6 Conclusion

We have presented a self-contained numerical study of Fourier-series convergence, the Gibbs phenomenon, and the discrete FFT analogue. All four studies reproduce the corresponding theoretical predictions to numerical precision. The distinction between the two Gibbs constants — $G_{\text{abs}} \approx 0.178980$ in absolute terms and $G_{\text{frac}} \approx 0.08949$ as a fraction of the jump — is emphasized, since the two are commonly conflated. The full implementation, three reproducible notebooks, and a test suite of ten unit tests are available in the accompanying repository.

Reproducibility

All figures and tables in this report were generated from the Jupyter notebooks in the accompanying repository:

<https://github.com/BrokerRecord/fourier-series-analysis>

The repository includes the full test suite (`pytest -v`, 10 tests, all passing) and a GitHub Actions workflow that runs on Python 3.10–3.12.

References

References

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- [5] P. Virtanen et al., *SciPy 1.0: fundamental algorithms for scientific computing in Python*, Nature Methods, 17:261–272, 2020.